Protein Language Models Improve Cold-Start Drug-Target Prediction

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Abstract

Generalizing to proteins never seen during training remains difficult for drug-target interaction models. We introduce PLM-DTI, which uses a pretrained protein language model (ESM-2) with contrastive learning to align drug and protein embeddings. We find that attention-based graph models do not generalize to unseen proteins, while protein language model features do. PLM-DTI improves cold-start AUC by 7% on the KIBA dataset.

1 Introduction

Most drug-target interaction benchmarks use random splits that leak protein information between training and test sets. In realistic discovery settings, the target protein is often novel. We show that cold-start evaluation exposes a large performance drop for graph-based models.

2 Method

PLM-DTI encodes proteins with the ESM-2 protein language model and drugs with a graph neural network. A contrastive learning objective pulls interacting drug-protein pairs together in a shared embedding space. We also compare against a Graph Attention Network baseline and a Transformer baseline.

3 Results

We evaluate on the DAVIS dataset and the KIBA dataset using cold-start protein splits. Results show that PLM-DTI reaches an AUC of 0.84 on cold-start KIBA, a 7% improvement over the Graph Attention Network baseline. The Graph Attention Network baseline drops from 0.90 AUC on random splits to 0.71 AUC on cold-start splits, indicating that attention-based graph models do not generalize to unseen proteins.

4 Limitations

PLM-DTI has a high computational cost because the protein language model requires large GPU memory. The approach does not use three-dimensional structural information about binding pockets. It remains unclear whether combining language model features with protein structure would further improve cold-start performance.

5 Conclusion

Protein language models provide transferable representations that substantially improve cold-start drug-target prediction.

References

[1] Placeholder reference for the synthetic paper.